A
Mini Project Report on

URL Phishing Detection And Email Spam Detection Using Machine Learning

Submitted in partial fulfillment of the requirements
for the degree of
BACHELOR OF ENGINEERING
IN
Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

by

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CERTIFICATE

This is to certify that the **"URL Phishing Detection And Email Spam Detection Using Machine Learning"** is a bonafide work of Pranet Pednekar (22106098), Dhruv Wesavkar (22106016), Akash Singh (22106165), Mohit Rajput (22106126) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning).**

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Project Report Approval

This Mini project report entitled "**URL Phishing Detection And Email Spam Detection Using Machine Learning**" by **Pranet Pednekar, Dhruv Wesavkar, Akash Singh and Mohit Rajput** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science &Engineering***, (AIML) ***2024-25***.

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission hasnot been taken when needed.

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ABSTRACT

Phishing attacks have become a significant cybersecurity threat, targeting individuals and organizations through deceptive emails, websites, and messages to steal sensitive information such as credentials, financial details, and personal data.

The proposed system leverages a combination of Gradient boosting classifier and feature extraction techniques to analyze URL structures, and uses XGBoost for email characteristics. A hybrid deep learning model is trained on a dataset of legitimate and phishing websites/emails to detect malicious patterns. Additionally, heuristic-based and blacklist-based methods are integrated to enhance detection accuracy. The system is designed to provide real-time phishing detection, ensuring user security and data protection. A web -based or browser-integrated tool can be deployed to alert users about potential threats before they interact with suspicious links. The primary goal of this project is to improve cybersecurity defenses by minimizing false positives while ensuring high detection rates, thereby reducing the risk of successful phishing attacks.

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CHAPTER 1

INTRODUCTION

1.INTRODUCTION

Cybersecurity threats are on the rise, with phishing attacks and email spam posing significant risks to individuals and organizations. Phishing websites deceive users into sharing sensitive information, while spam emails serve as a medium for scams, malware distribution, and fraudulent schemes. To combat these threats, machine learning models offer a powerful solution by identifying suspicious URLs and filtering spam emails with high accuracy.

This project focuses on two major aspects of cybersecurity: Phishing URL Detection and Email Spam Detection. For Phishing URL Detection, we employ a Gradient Boosting Model (GBM), a robust ensemble learning technique that improves classification performance by sequentially reducing errors. The model is trained on various URL-based features, such as domain age, presence of special characters, and HTTP/HTTPS usage, to distinguish between legitimate and malicious websites effectively.

For Email Spam Detection, we implement an XGBoost model, a high-performance gradient boosting algorithm that excels in handling large datasets and improving classification accuracy. The model is trained on a dataset of email texts, utilizing Natural Language Processing (NLP) techniques such as TF-IDF (Term Frequency-Inverse Document Frequency) and word embeddings to extract meaningful patterns. The goal is to classify emails as either spam or legitimate based on their content and metadata.

Both models are evaluated using performance metrics like accuracy, precision, recall, and F1-score to ensure reliable detection. The project not only enhances cybersecurity awareness but also provides a scalable solution for automated threat detection. By integrating these models into real-world applications, users can safeguard themselves against fraudulent activities, ensuring a safer online experience.

CHAPTER 2

LITERATURE SURVEY

2.LITERATURE SURVEY

2.1-HISTORY

Phishing attacks have been a growing cybersecurity concern since the early days of the internet. The term "phishing" was first used in the mid-1990s when attackers began using fraudulent emails and websites to steal sensitive information from users. Over the years, phishing techniques have evolved, becoming more sophisticated and harder to detect.

1. Early Phishing Attacks (1990s - Early 2000s):

- The first phishing attacks targeted users of America Online (AOL) in the mid-1990s, where attackers used fake login pages to steal credentials.
- Email-based phishing became widespread in the early 2000s, often impersonating banks and financial institutions to trick users into providing sensitive information.

2. Rise of Anti-Phishing Measures (Mid-2000s - 2010s):

- Blacklist-based methods were introduced, where known phishing URLs were stored in databases to warn users.
- Email providers and web browsers implemented spam filters and warning systems to detect suspicious links.
- The Anti-Phishing Working Group (APWG) was formed to combat phishing threats by sharing intelligence among cybersecurity experts.

3. Machine Learning and AI-Based Detection (2010s - Present):

- With the rise of AI, phishing detection systems started using machine learning algorithms to analyze website URLs, text patterns, and metadata.
- Natural Language Processing (NLP) and deep learning models improved detection accuracy by identifying phishing attempts based on text structure and behavioral patterns.
- Real-time phishing detection tools, such as Google Safe Browsing and Microsoft SmartScreen, became widely adopted to warn users before accessing suspicious sites.

The continuous evolution of phishing attacks highlights the need for advanced, automated detection systems. Modern phishing detection solutions leverage AI and real-time threat intelligence to provide proactive protection, ensuring users and organizations stay ahead of cyber threats.

2.2-LITERATURE REVIEW

[1] Shaik, Hasane. (2022). Phishing URL detection using machine learning methods. 103288.

Phishing is a major cybercrime that exploits URLs to steal user data. Detecting malicious URLs is challenging, as many evade blacklists due to recentness or misclassification. This work proposes a machine learning-based solution using algorithms like Random Forests, Decision Trees, LightGBM, Logistic Regression, and SVM. By extracting key features from URLs, the model aims to enhance web security and improve detection accuracy.

[2] A. Alswailem, B. Alabdullah, N. Alrumayh and A. Alsedrani, "Detecting Phishing Websites Using Machine Learning," 2019 2nd International Conference on Computer Applications & Information Security (ICCAIS), Riyadh, Saudi Arabia, 2019, pp. 1-6, doi: 10.1109/CAIS.2019.8769571.

Phishing websites exploit human vulnerabilities to steal sensitive information. This paper presents a browser extension that detects phishing sites using supervised machine learning. The system employs the Random Forest algorithm for its strong classification performance. By selecting an optimal combination of 26 features, our model achieves 98.8% accuracy in phishing detection.

[3] A. Mandadi, S. Boppana, V. Ravella and R. Kavitha, "Phishing Website Detection Using Machine Learning," 2022 IEEE 7th International conference for Convergence in Technology (I2CT), Mumbai, India, 2022, pp. 1-4, doi: 10.1109/I2CT54291.2022.9824801.

Phishing is an internet scam where attackers send fake messages to steal personal data or spread malware. Machine learning is an effective approach to detect phishing attempts by analyzing URLs. This study employs Random Forest and Decision Tree classifiers, achieving 87.0% and 82.4% accuracy, respectively, in distinguishing phishing from legitimate URLs.

[4] S. Parekh, D. Parikh, S. Kotak and S. Sankhe, "A New Method for Detection of Phishing Websites: URL Detection," 2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT), Coimbatore, India, 2018, pp. 949-952, doi: 10.1109/ICICCT.2018.8473085.

The primary focus of this paper is to put forth a model as a solution to detect phishing websites

by using the URL detection method using Random Forest algorithm. There are 3 major phases such as Parsing, Heuristic Classification of data, Performance Analysis in this model and each phase makes use of a different technique or algorithm for processing of data to give better results.

[5] A. Chien and P. Khethavath, "Email Feature Classification and Analysis of Phishing Email Detection Using Machine Learning Techniques," 2023 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE), Nadi, Fiji, 2023, pp. 1-8, doi: 10.1109/CSDE59766.2023.10487729.

This paper introduced new features that were identified in modern emails and conducted two types of experiments using machine learning techniques. The first experiment focused on how the extracted features weighted the importance when classifying legitimate advertisements and phishing emails. And the second experiment focused on legitimate emails and phishing emails. We also evaluated these machine learning algorithms on a real dataset that we gathered, and some of the results were below par when compare with prefetched datasets.

[6] N. Kumar, S. Sonowal and Nishant, "Email Spam Detection Using Machine Learning Algorithms," ***2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA)*****, Coimbatore, India, 2020, pp. 108-113, doi: 10.1109/ICIRCA48905.2020.9183098.**

Email Spam has become a major problem nowadays, with Rapid growth of internet users, Email spams is also increasing. People are using them for illegal and unethical conducts, phishing and fraud. Sending malicious link through spam emails which can harm our system and can also seek in into your system. Creating a fake profile and email account is much easy for the spammers, they pretend like a genuine person in their spam emails, these spammers target those peoples who are not aware about these frauds. So, it is needed to Identify those spam mails which are fraud, this project will identify those spam by using techniques of machine learning, this paper will discuss the machine learning algorithms and apply all these algorithm on our data sets and best algorithm is selected for the email spam detection having best precision and accuracy.

[7] S. Nandhini and J. Marseline K.S., "Performance Evaluation of Machine Learning Algorithms for Email Spam Detection," ***2020 International Conference on Emerging***

***Trends in Information Technology and Engineering (ic-ETITE)*, Vellore, India, 2020, pp. 1-4, doi: 10.1109/ic-ETITE47903.2020.312.**

This paper discusses the efficient methods of using some of the popular algorithms for building a machine learning model which can classify whether a mail is a spam or ham. UCI Machine Learning Repository Spambase Data Set is used for the experiment. The performance of five important machine learning classification algorithms viz. Logistic Regression, Decision Tree, Naive Bayes, KNN and SVM are evaluated in order to train and build an effective machine learning model for email spam detection. Weka tool is used for training and testing the data set.

[8] Opeyemi Emmanuel Ajibuwa, Emmanuel Gbenga Dada, Joseph Stephen Bassi, Haruna Chiroma, Adebayo Olusola Adetunmbi and Shafi'i Muhammad Abdulhamid, "Machine learning for email spam filtering: review, approaches and open research problems", doi: 10.1016/j.heliyon.2019.e01802

We present a systematic review of some of the popular machine learning based email spam filtering approaches. Our review covers survey of the important concepts, attempts, efficiency, and the research trend in spam filtering. The preliminary discussion in the study background examines the applications of machine learning techniques to the email spam filtering process of the leading internet service providers (ISPs) like Gmail, Yahoo and Outlook emails spam filters.

[9] Isra'a AbdulNabi, Qussai Yaseen, "Spam Email Detection Using Deep Learning Techniques", Procedia Computer Science, Volume 184, 2021, Pages 853-858, ISSN 1877-0509, doi: 10.1016/j.procs.2021.03.107.

In this work, the effectiveness of word embedding in classifying spam emails is introduced. Pre-trained transformer model BERT (Bidirectional Encoder Representations from Transformers) is fine-tuned to execute the task of detecting spam emails from non-spam (HAM). BERT uses attention layers to take the context of the text into its perspective. Results are compared to a baseline DNN (deep neural network) model that contains a BiLSTM (bidirectional Long Short Term Memory) layer and two stacked Dense layers. In addition results are compared to a set of classic classifiers k-NN (k-nearest neighbors) and NB (Naive Bayes). Two open-source data sets are used, one to train the model and the other to test the persistence and robustness of the model against unseen data. The proposed approach attained

the highest accuracy of 98.67% and 98.66% F1 score.

[10] A. Karim, M. Shahroz, K. Mustofa, S. B. Belhaouari and S. R. K. Joga, "Phishing Detection System Through Hybrid Machine Learning Based on URL," in *IEEE Access*, vol. 11, pp. 36805-36822, 2023, doi: 10.1109/ACCESS.2023.3252366.

The proposed study is based on the phishing URL-based dataset extracted from the famous dataset repository, which consists of phishing and legitimate URL attributes collected from 11000+ website datasets in vector form. After preprocessing, many machine learning algorithms have been applied and designed to prevent phishing URLs and provide protection to the user. This study uses machine learning models such as decision tree (DT), linear regression (LR), random forest (RF), naive Bayes (NB), gradient boosting classifier (GBM), K-neighbors classifier (KNN), support vector classifier (SVC), and proposed hybrid LSD model, which is a combination of logistic regression, support vector machine, and decision tree (LR+SVC+DT) with soft and hard voting, to defend against phishing attacks with high accuracy and efficiency. The canopy feature selection technique with cross fold valoidation and Grid Search Hyperparameter Optimization techniques are used with proposed LSD model. Furthermore, to evaluate the proposed approach, different evaluation parameters were adopted, such as the precision, accuracy, recall, F1-score, and specificity, to illustrate the effects and efficiency of the models. The results of the comparative analyses demonstrate that the proposed approach outperforms the other models and achieves the best results.

CHAPTER 3

PROBLEM STATEMENT

3. Problem Statement

Phishing attacks have become one of the most common and dangerous cybersecurity threats, targeting individuals and organizations to steal sensitive information such as login credentials, banking details, and personal data. Cybercriminals use deceptive emails, fake websites, and social engineering tactics to trick users into providing confidential information. Traditional phishing detection methods, such as blacklist-based approaches and rule-based systems, are ineffective against newly generated phishing URLs and sophisticated obfuscation techniques.

This project aims to develop an advanced phishing detection system using machine learning techniques to analyze and classify URLs, emails, and website content in real-time. The system will extract relevant features, such as domain characteristics, text patterns, and structural anomalies, to identify phishing attempts with high accuracy. By integrating artificial intelligence, the proposed solution seeks to enhance cybersecurity by providing users with a reliable and efficient mechanism to detect and prevent phishing attacks before they cause harm.

CHAPTER 4
EXPERIMENTAL SETUP

4. Experimental Setup

4.1 Hardware Setup

- CPU: Intel Core i7/i9
- GPU: NVIDIA GTX 3060/3070/4090
- RAM: 16 GB
- Storage: 512 GB SSD

4.2 Software Setup

The following tools or framework were used:

- HTML/CSS
- Python 3.9
- beautifulsoup4
- googlesearch_python
- Numpy
- Pandas
- python_dateutil
- Requests
- scikit_learn
- urllib3
- whois
- gunicorn

CHAPTER 5
PROPOSED SYSTEM AND IMPLEMENTATION

5. Proposed System & Implementation

5.1 BLOCK DIAGRAM OF PROPOSED SYSTEM

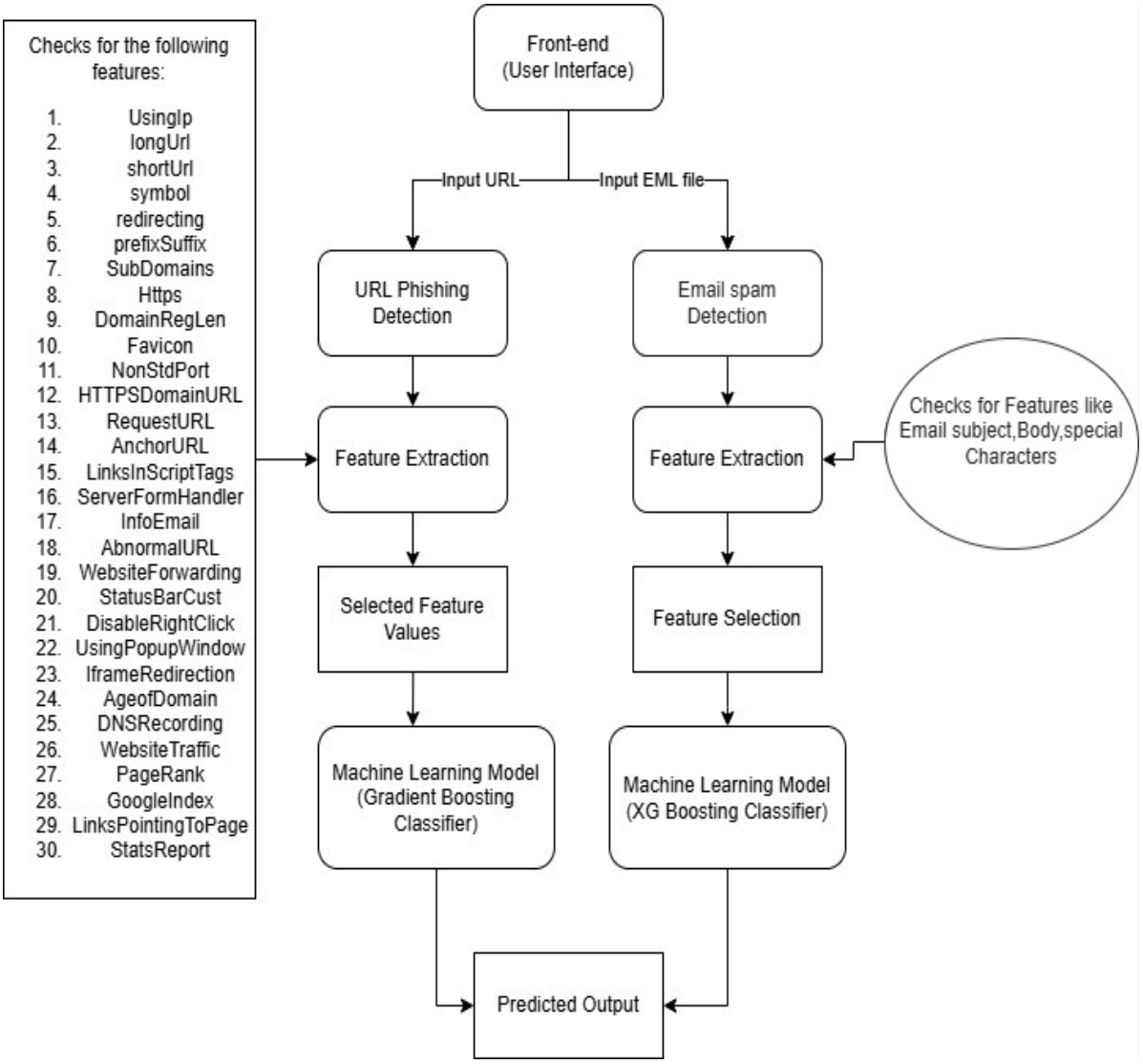

Fig 5.1.1 Block Diagram of Proposed System

5.2 DESCRIPTION OF BLOCK DIAGRAM

The block diagram represents a phishing detection system with two main modules: URL phishing detection and email phishing detection.

1. User Input: A user provides either a URL or an email (EML file) through the front-end interface.

2. Feature Extraction:

- For URL phishing detection, 30 features are checked, such as URL length, presence of special characters, and redirections.

- For email phishing detection, features like the number of URLs and special characters are extracted.

3. Feature Processing: Selected features are fed into machine learning models.

4. Classification:

- A Gradient Boosting Classifier predicts phishing URLs.
- An XGBoost Classifier identifies phishing emails.

5. Output: The system provides the predicted result, indicating whether the input is phishing or not.

5.3 IMPLEMENTATION

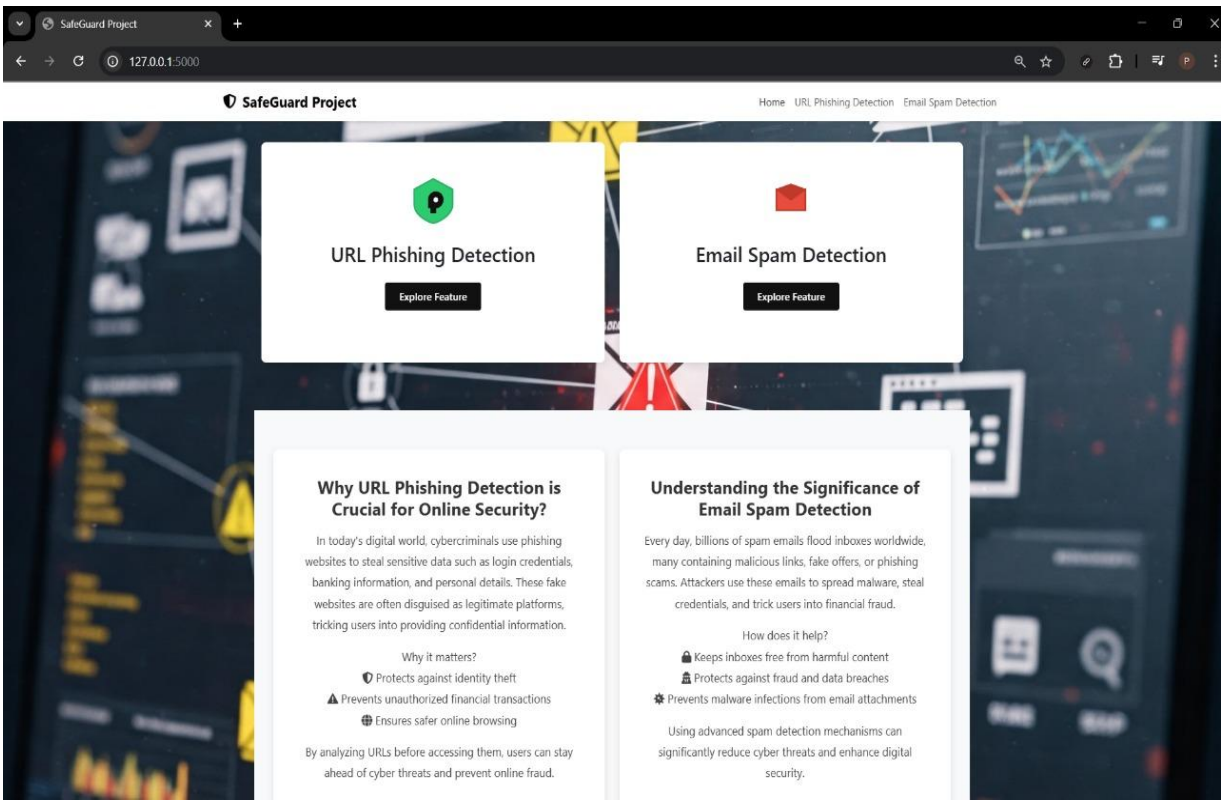

Fig 5.3.1 Homepage of Developed System

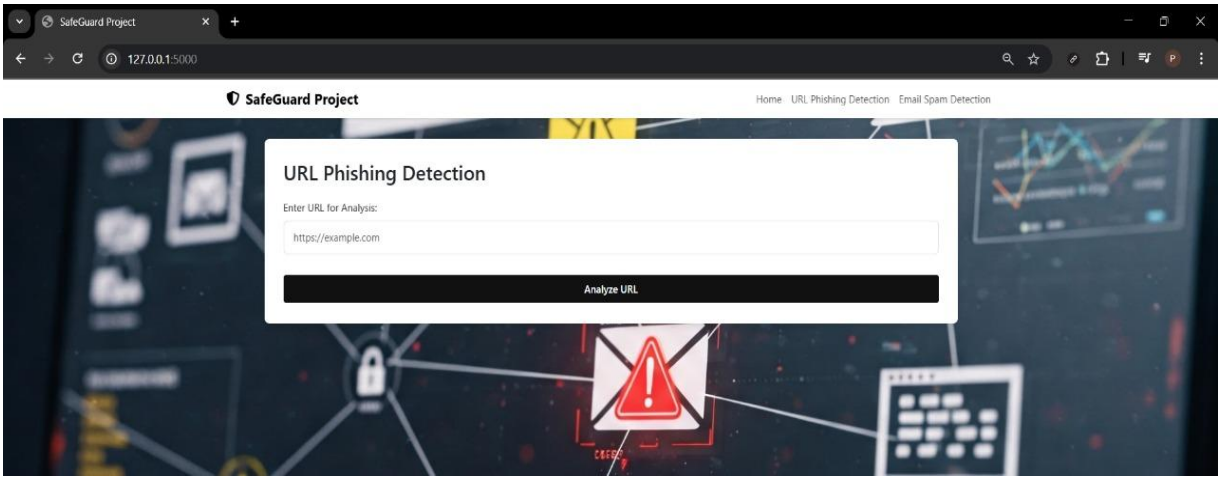

Fig 5.3.2 URL Phishing Detection Page

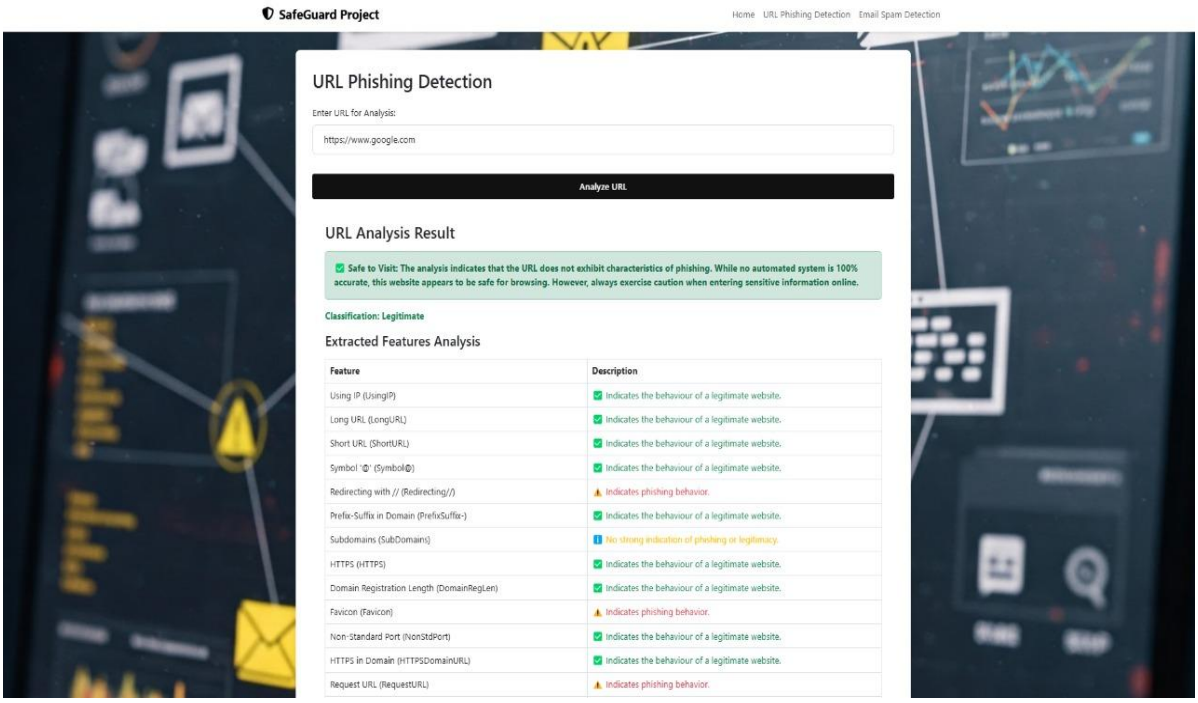

Fig 5.3.3 SafeLink Analysis Results Dashboard

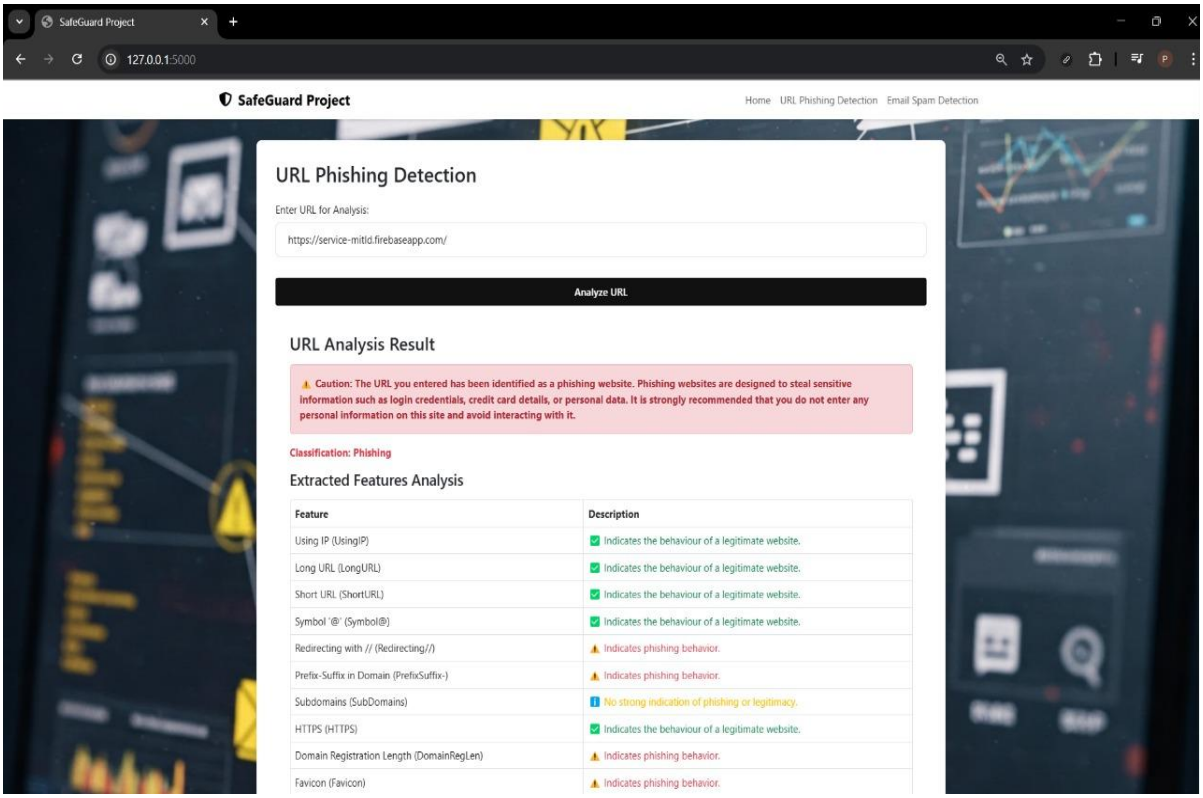

Fig 5.3.4 SafeLink Phishing URL Analysis Results

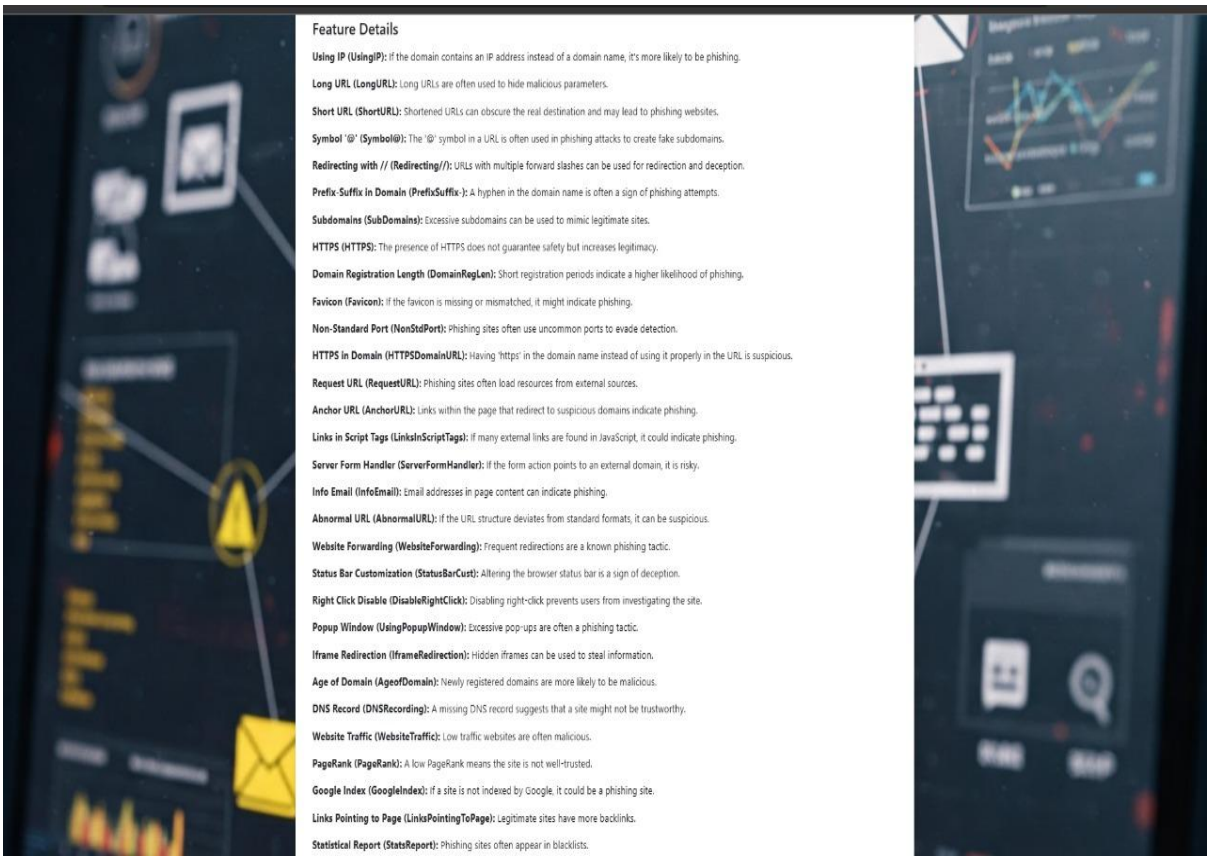

Fig 5.3.5 Feature Details of URL

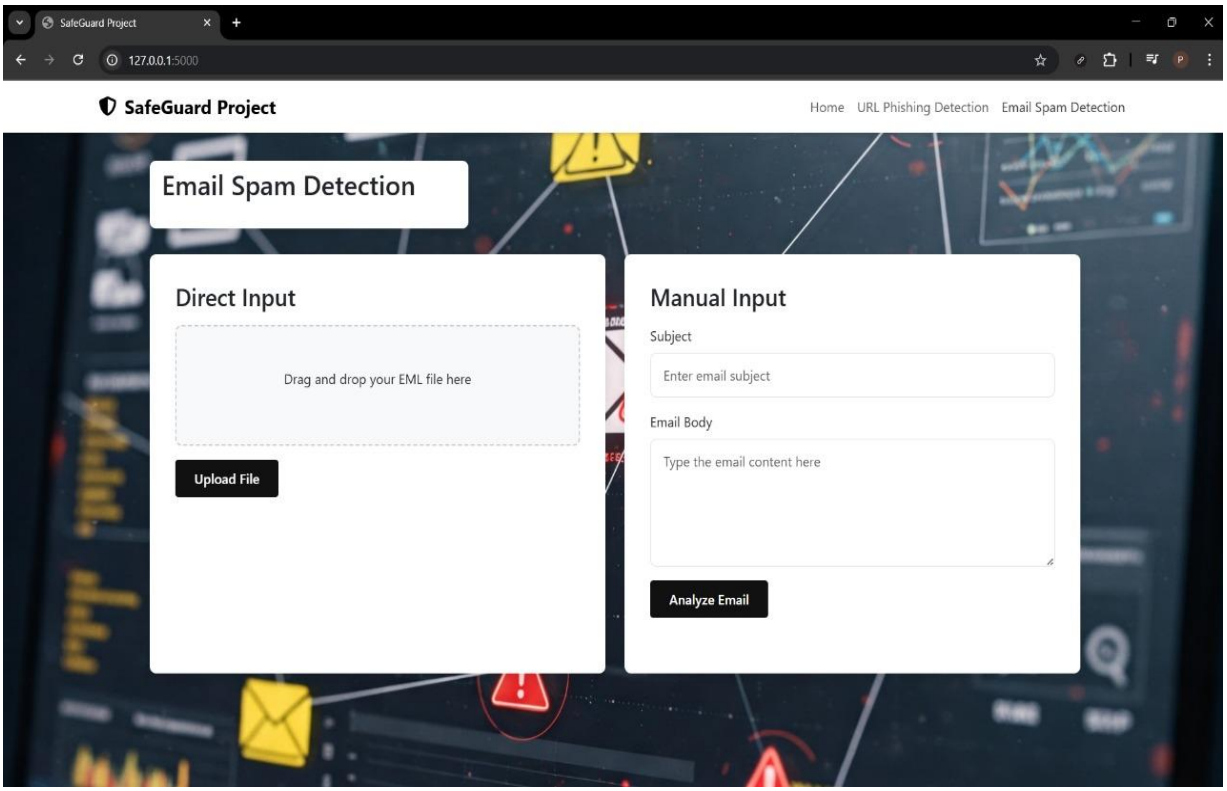

Fig 5.3.6 Email Spam Detection Page

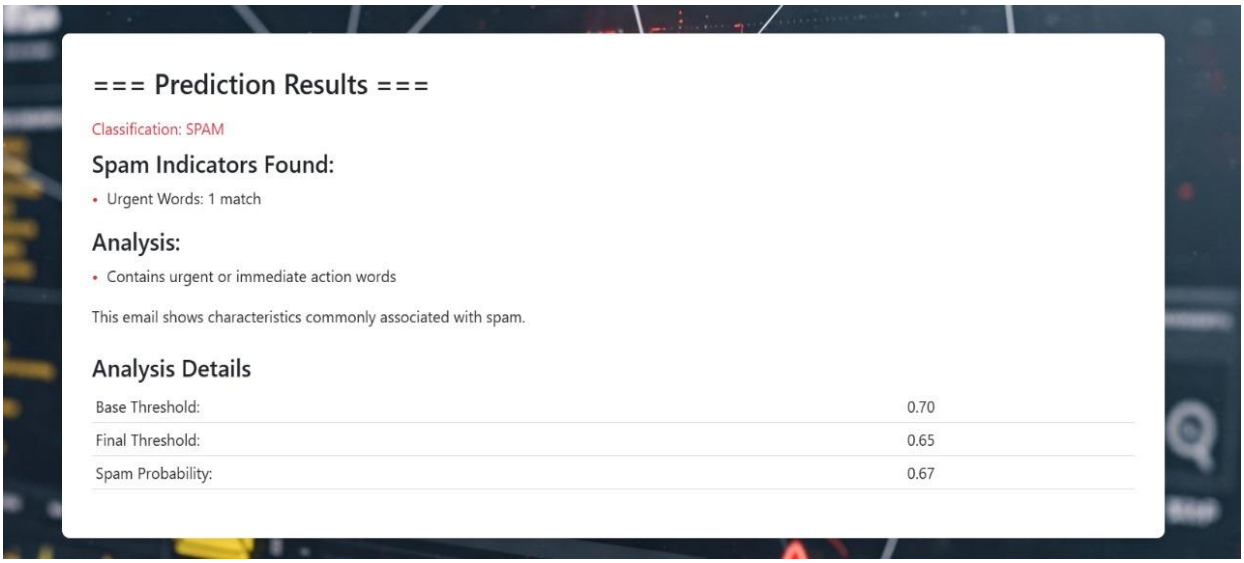

Fig 5.3.7 Prediction Results for Email

5.4 ADVANTAGES / APPLICATION / RESULT TABLE

Analysis of Model Statistics for URL Phishing Detection:

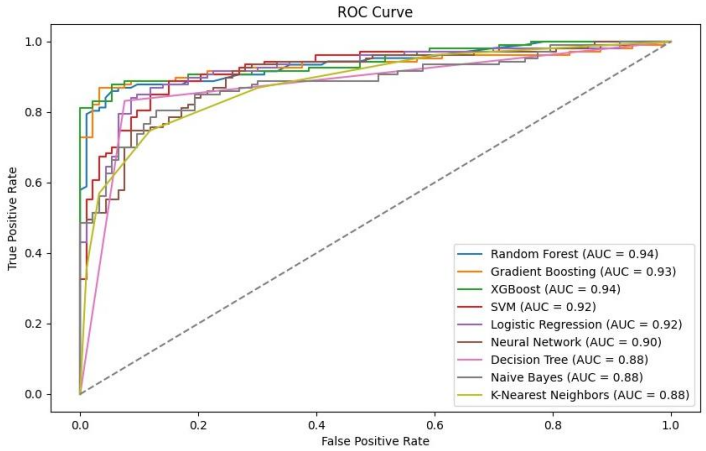

Fig 5.4.1 ROC Curve of URL Model

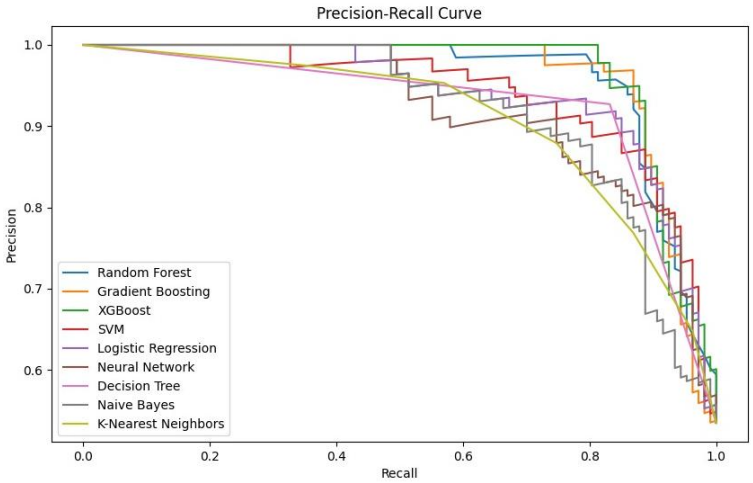

Fig 5.4.2 Precision-Recall Curve of URL Model

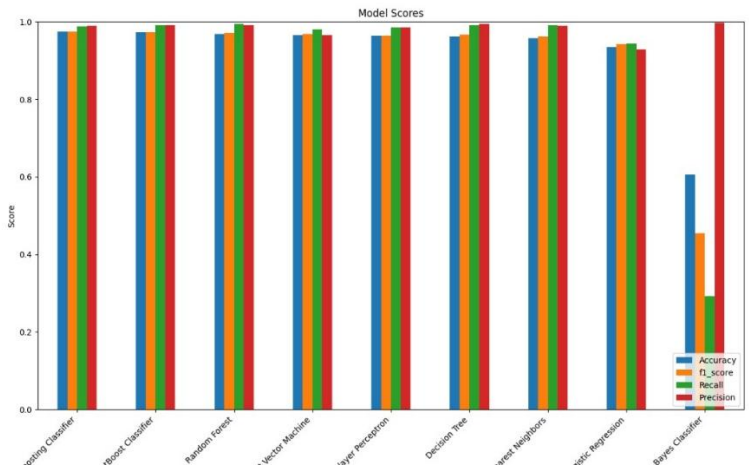

Fig 5.4.3 Performance Metrics Comparison of URL Model

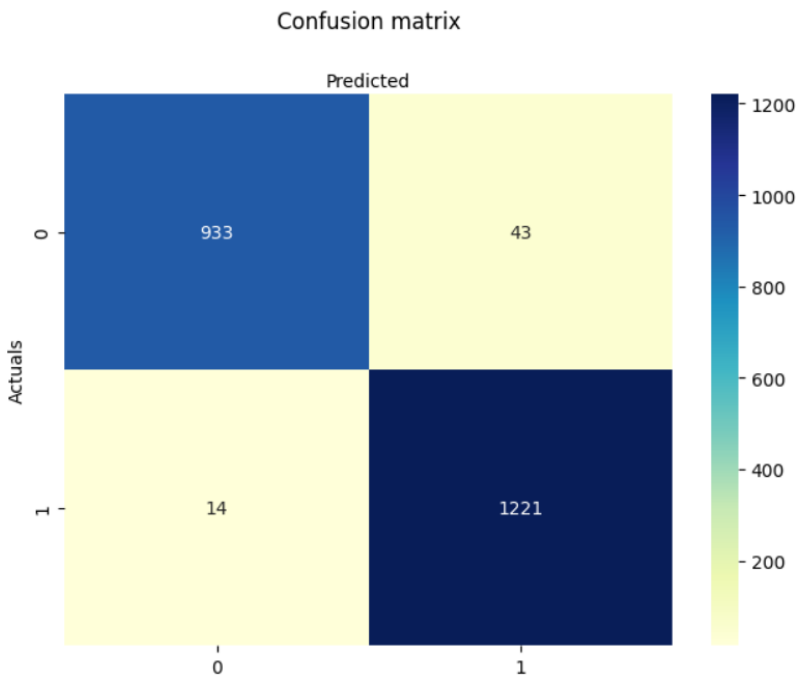

Fig 5.4.4 Confusion Matrix of URL Model

The ROC Curve and Precision-Recall Curve indicate the overall performance of different machine learning models. The Gradient Boosting Model (AUC=0.93) shows competitive performance compared to Random Forest (AUC=0.94) and XGBoost (AUC=0.94). The bar chart comparing accuracy, precision, recall, and F1-score highlights that Gradient Boosting performs consistently well across all metrics, making it a reliable choice.

We chose Gradient Boosting because it balances high accuracy, precision, and recall, ensuring fewer false positives and false negatives. While Random Forest and XGBoost also perform well, Gradient Boosting is computationally efficient, reducing variance and

improving prediction reliability. Additionally, it is robust to imbalanced data, a crucial factor in phishing detection where malicious URLs are often rare. Therefore, Gradient Boosting offers an optimal trade-off between performance and computational efficiency, making it a preferred choice.

Analysis of Model Statistics for E-Mail Spam Detection:

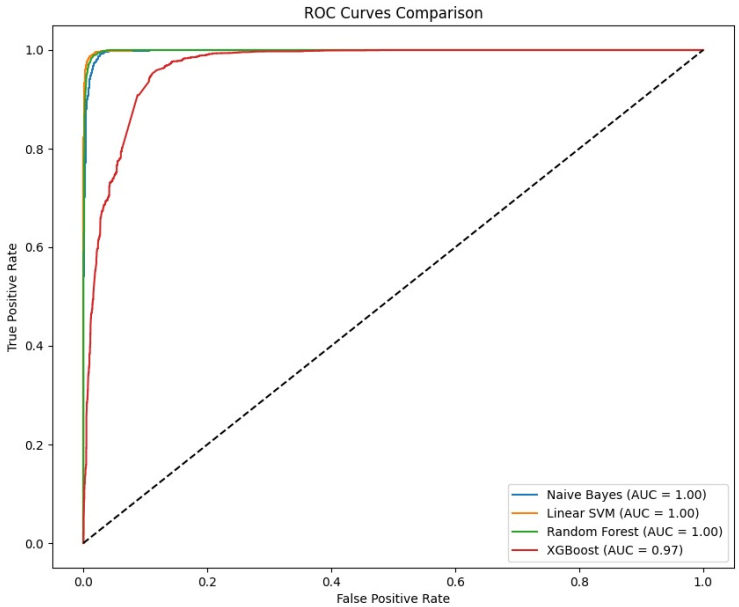

Fig 5.4.5 ROC Curve of E-Mail Spam Detection

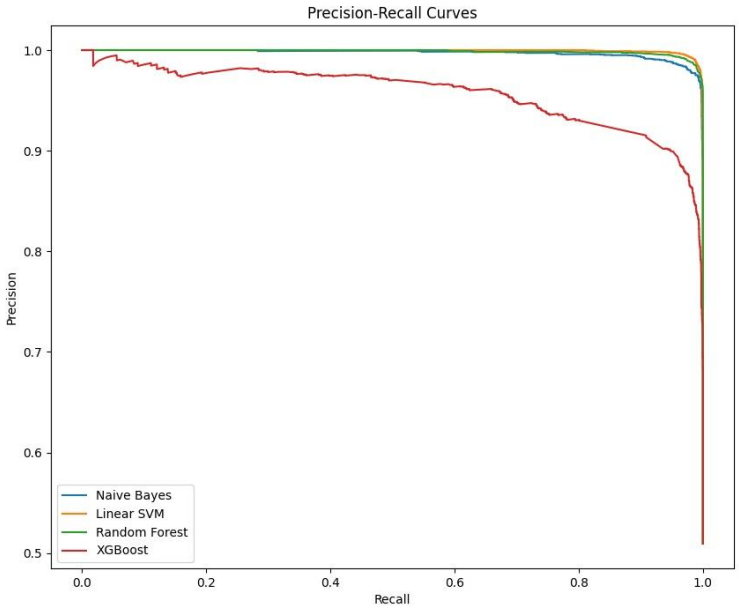

Fig 5.4.6 Precision-Recall Curve of E-Mail Spam Detection

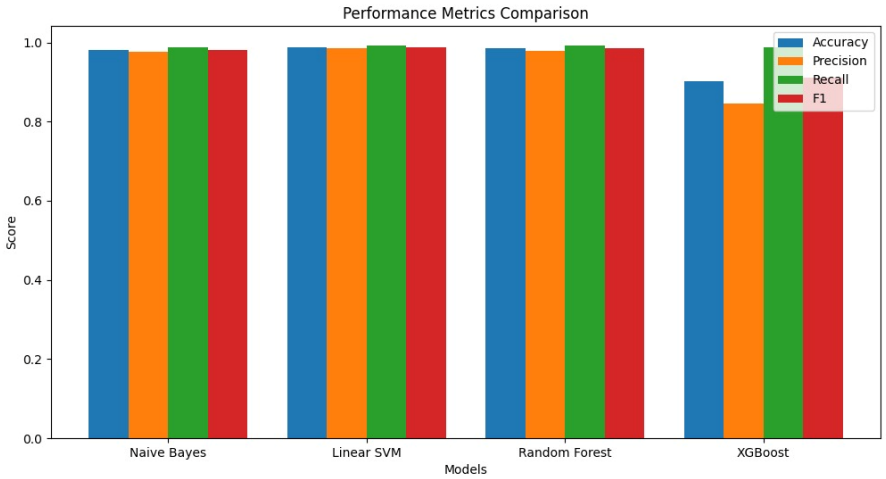

Fig 5.4.7 Performance Metrics Comparison of E-Mail Spam Detection

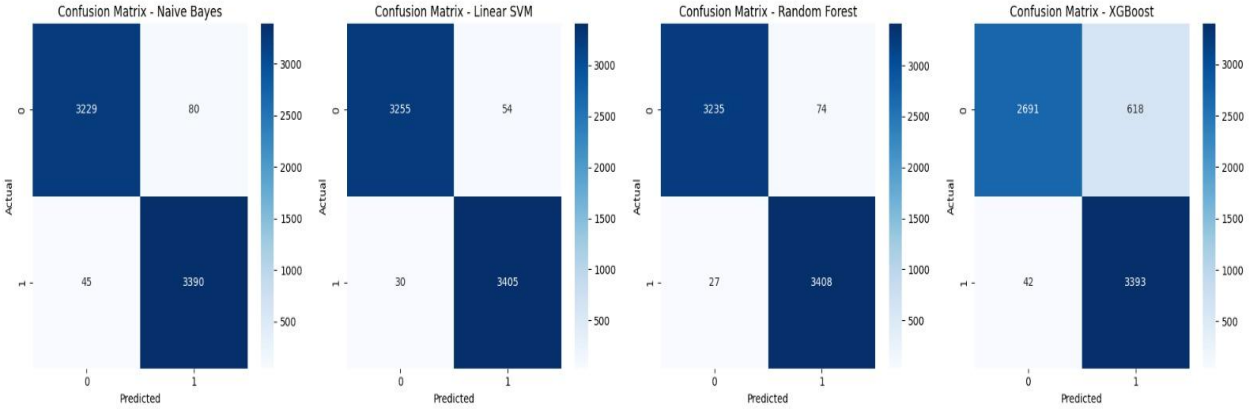

Fig 5.4.8 Confusion Matrix of E-Mail Spam Detection

The ROC and Precision-Recall curves indicate that Naïve Bayes, Linear SVM, and Random Forest achieve an AUC of 1.00, while XGBoost achieves 0.97. The performance metrics chart shows that Naïve Bayes, SVM, and Random Forest have slightly higher accuracy, precision, recall, and F1-scores than XGBoost. However, the confusion matrix reveals an important detail—XGBoost maintains a strong balance between false positives and false negatives, even though it predicts more false positives than other models.

Why XGBoost?

Despite its slightly lower AUC, XGBoost provides excellent generalization while handling class imbalances effectively. It is known for efficient computation, scalability, and robustness to noise, making it ideal for real-world spam filtering. Additionally, Naïve Bayes assumes feature independence, which may not always hold in spam detection, while SVM and Random Forest can be computationally expensive. Thus, XGBoost is chosen for its trade-off between performance and efficiency.

CHAPTER 6
CONCLUSION

Conclusion

In this project, we successfully developed and evaluated machine learning models for URL Phishing Detection and Email Spam Detection using Gradient Boosting and XGBoost, respectively. Through extensive experimentation, we compared multiple models, including Random Forest, SVM, Naïve Bayes, and Neural Networks, using key performance metrics such as AUC-ROC, Precision-Recall, Accuracy, Precision, Recall, and F1-score.

For URL Phishing Detection, Gradient Boosting was chosen due to its high accuracy, robust feature learning, and ability to handle imbalanced data effectively. For Email Spam Detection, XGBoost emerged as the optimal choice, offering strong generalization, efficient computation, and the ability to manage complex patterns in textual data.

Overall, this project demonstrates the power of machine learning in enhancing cybersecurity by automating threat detection with high precision and recall. Future enhancements could include deep learning models, ensemble techniques, and real-time deployment for dynamic threat prevention.

References

Research paper

[1] Shaik, Hasane. (2022). Phishing URL detection using machine learning methods. 103288.

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